EML 3022: CAD Project FALL 2024

ZPU - 4

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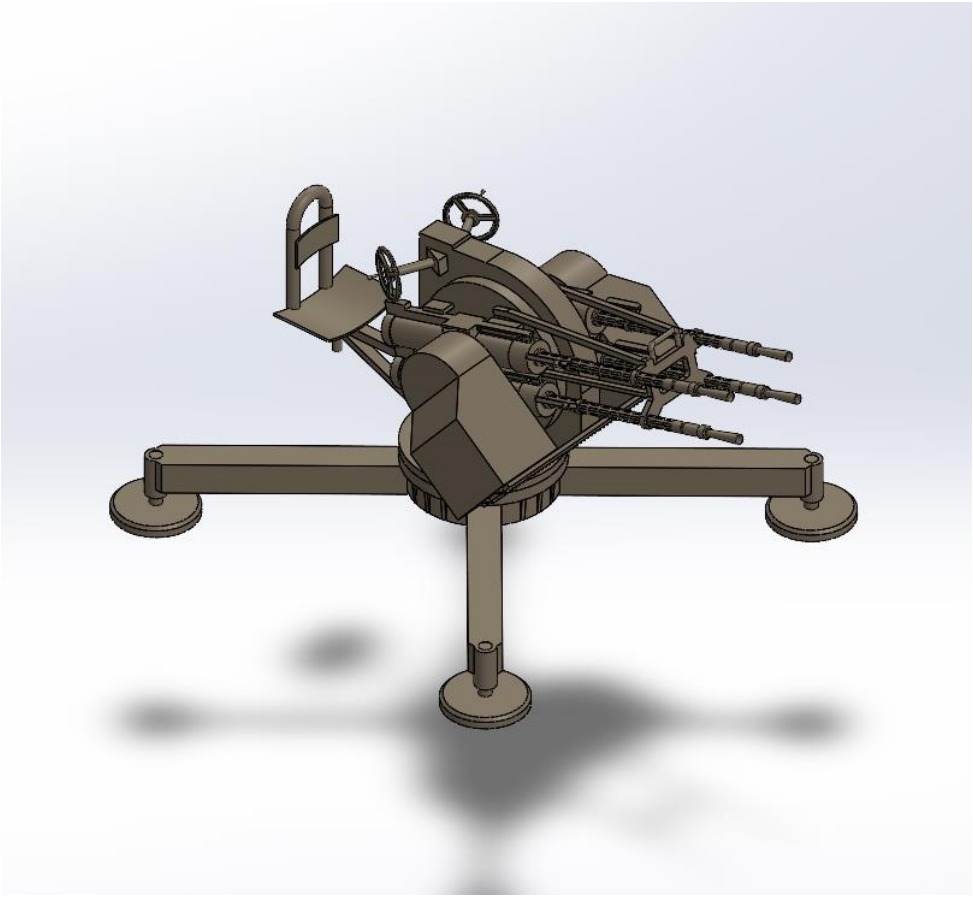


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Introduction:

The design the I chose to make is the Russian-made anti-aircraft gun called the ZPU-4. The application of this machine is to take down enemy aircraft. I was told that we should choose a challenging design to create so I was thinking along the lines of weaponry and then got to the idea of an anti-aircraft gun. I was thinking about the game called Battlefield 4 and was thinking about one of my favorite vehicles in the game. It was called the " 2/**9K22 Tunguska**," but I didn't know that and so I just searched up "Russian anti-aircraft guns," and found a list of said guns. I was looking through images and came across these cool looking 4 barrels stationary guns and was like, OK I am going to design this.

The Design:

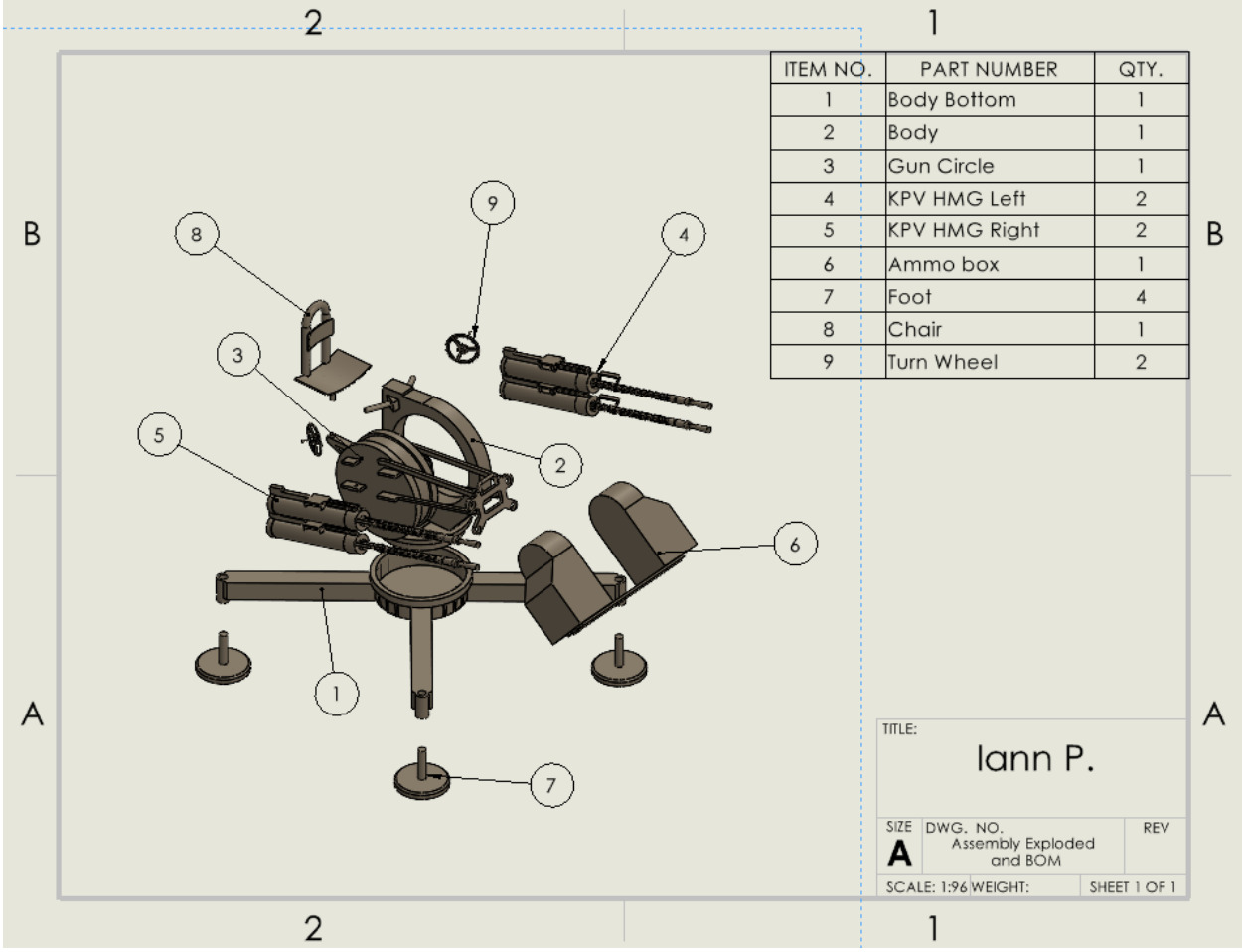
Materials include:

- Cast Alloy Steel – I was thinking, since these were massed produced, common material would be used. This is for the body of the ZPU-4
- Plain Carbon Steel – Needed stronger materials for the rotating components for the ZPU – 4, also I made the guns out of this material too.
- Military Tan paint – I used this paint because it had to camouflage in desert warfare.

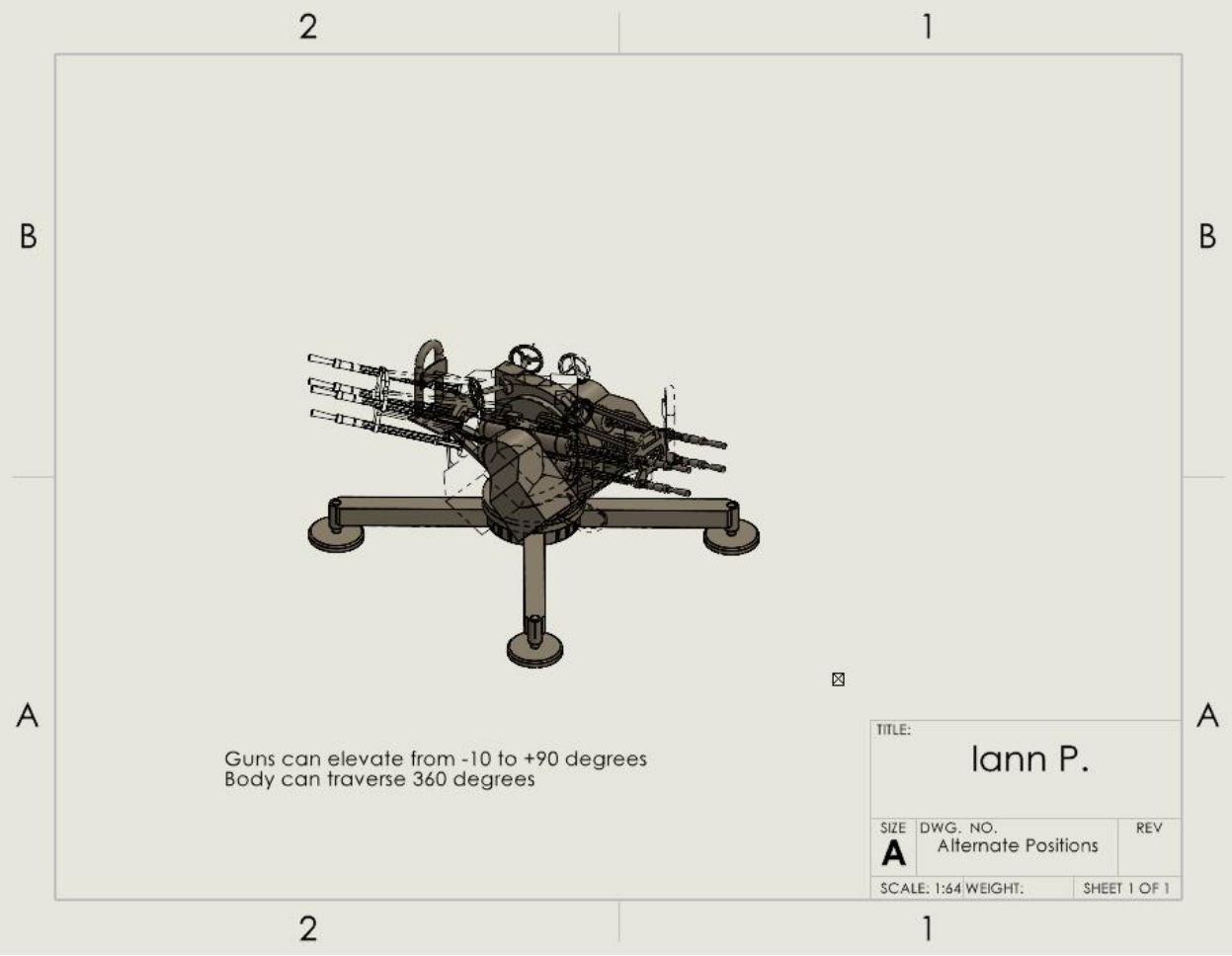
Difficulties:

The difficulties that came with this project was trying to figure out what were the correct dimensions for this ZPU-4. There would be websites that would give dimensions of the length, width and height, but when cross-referencing with other websites, there would be different values, and it wasted a lot of time trying to figure out exact dimensions. Ultimately, I was able to figure out the dimensions for the machine guns equipped on the ZPU-4 and I based every other component according to the machine guns. What was key was having reference photos of the ZPU – 4, I had many reference photos from the web and even top, right and front views to work with for this project. I had to learn how to put the photos into the planes in SolidWorks, from there I scaled the reference photos to match the size of the guns and then that's how I made everything else. Originally, I was going to make a moving ZPU – 4 that could be towed like you see in the images, but because of the different dimensions given by different websites, the scaling was very off and so I just decided to just make it a stationary ZPU – 4. **DIMENSIONS SHOWN BELOW ARE FROM EYEBALLING.**

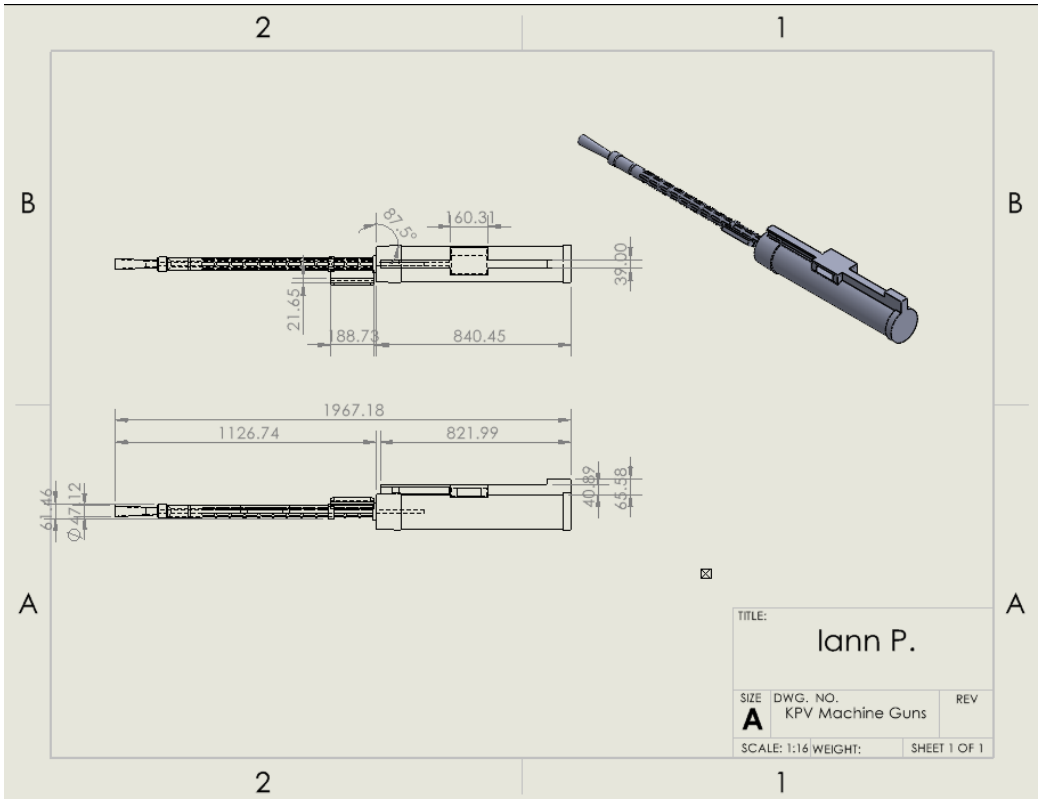
Assembly (Exploded)

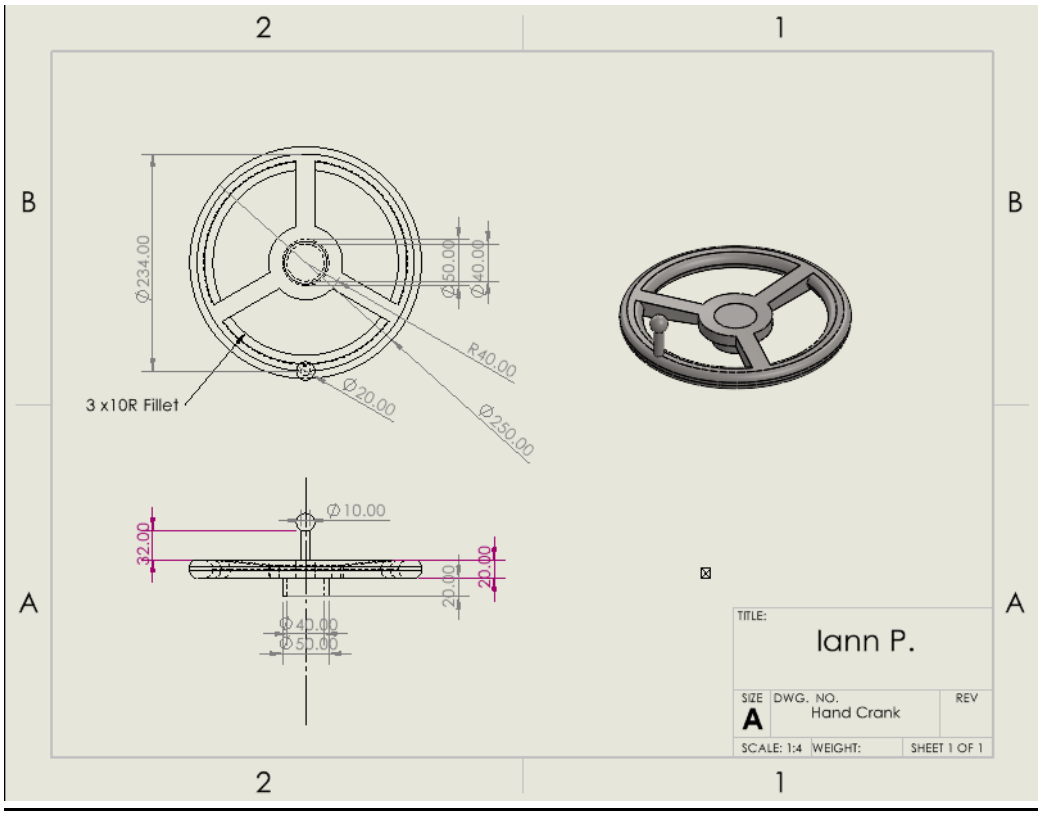


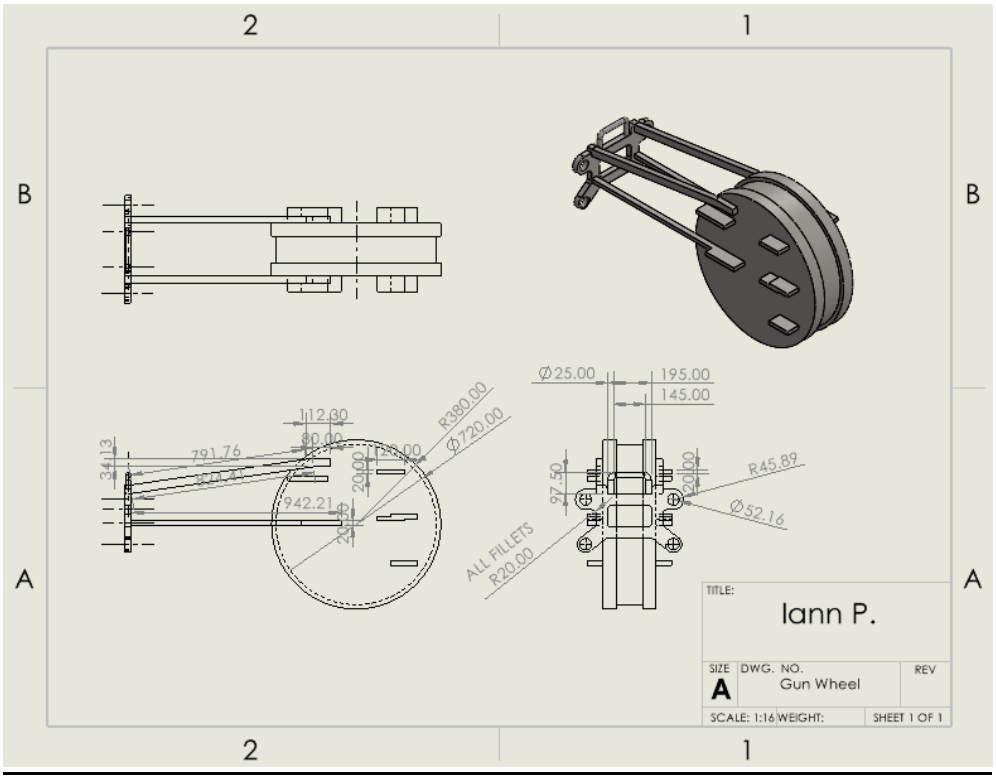
Assembly (Alternate Positions)

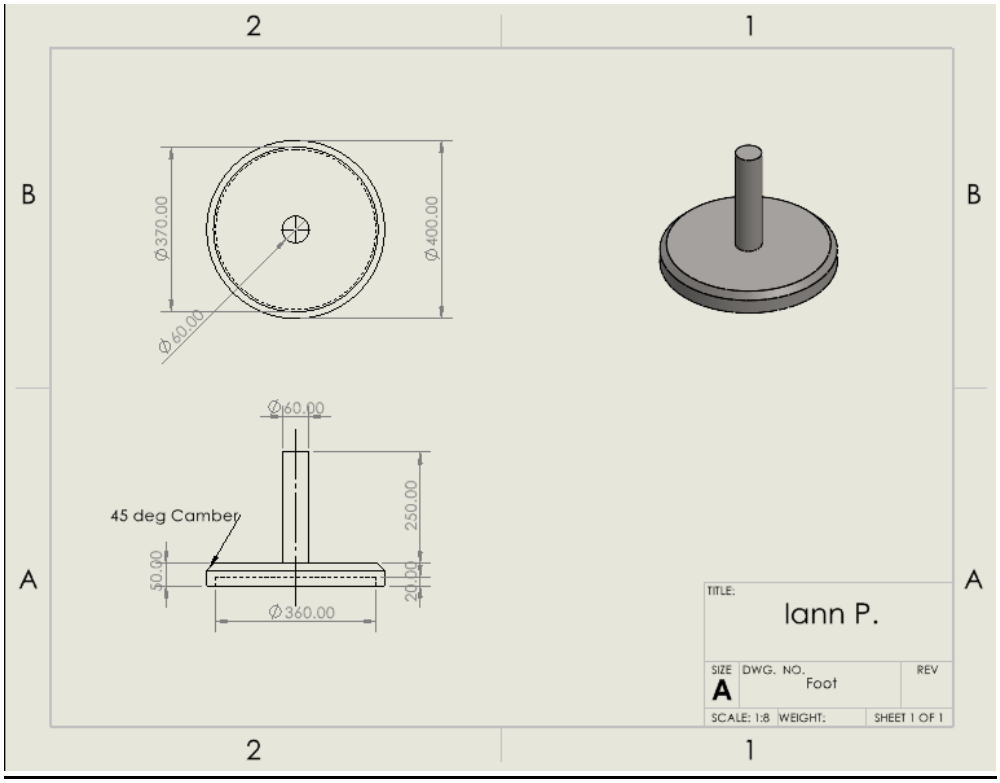


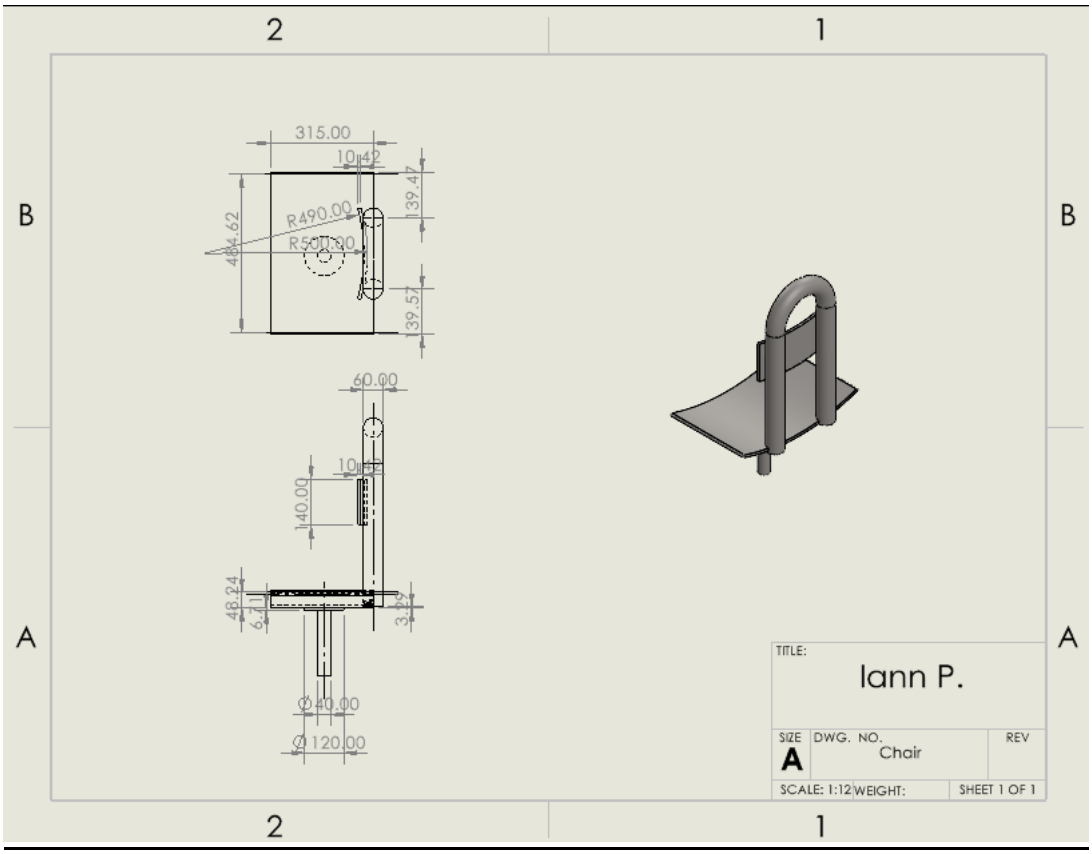
Part Drawings

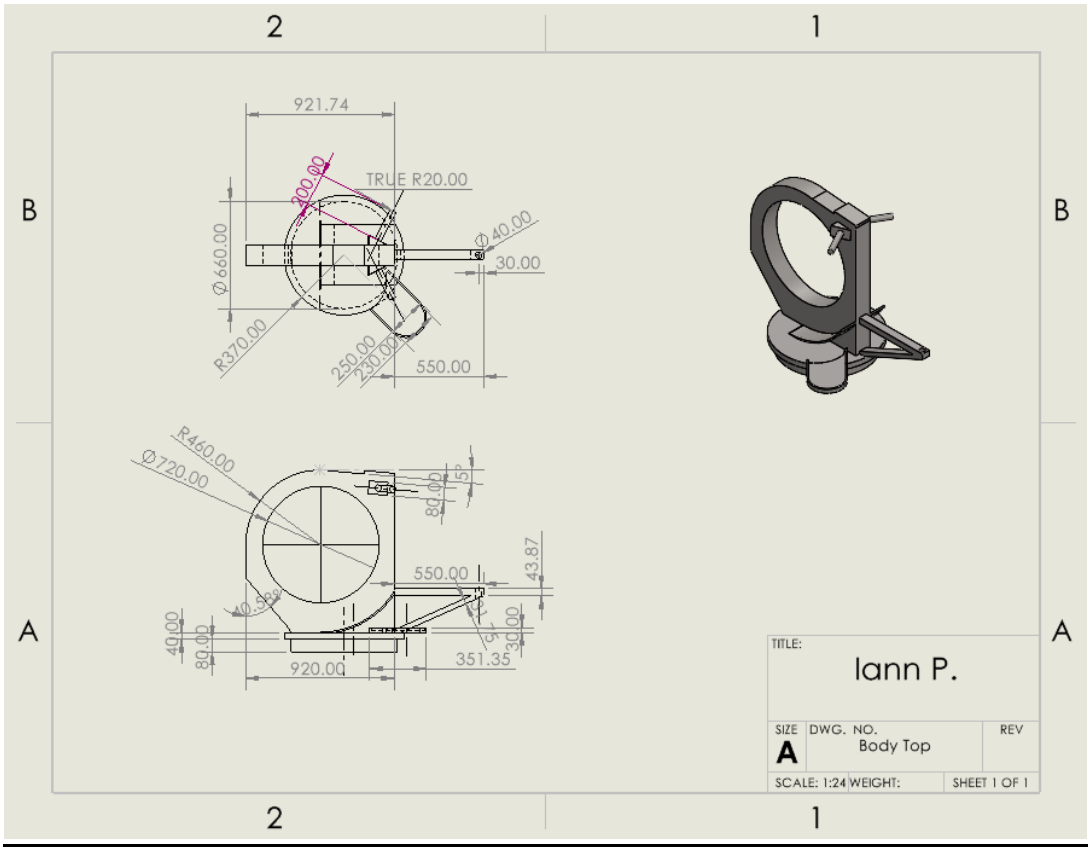


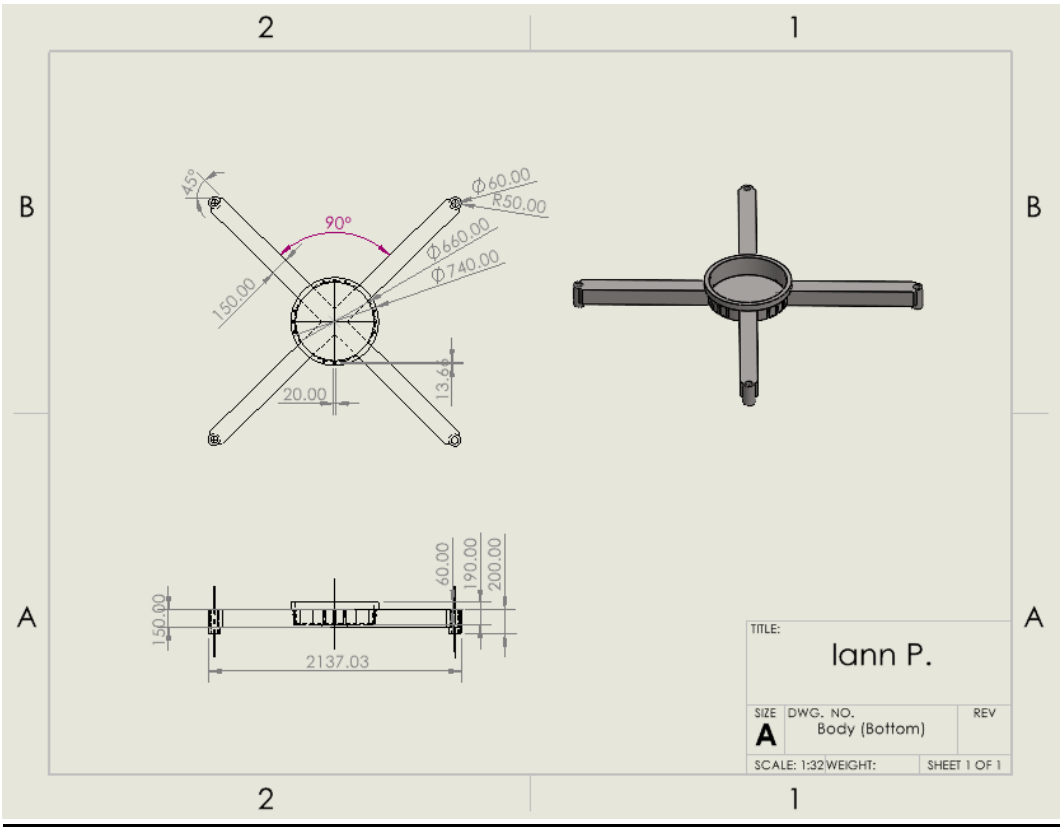


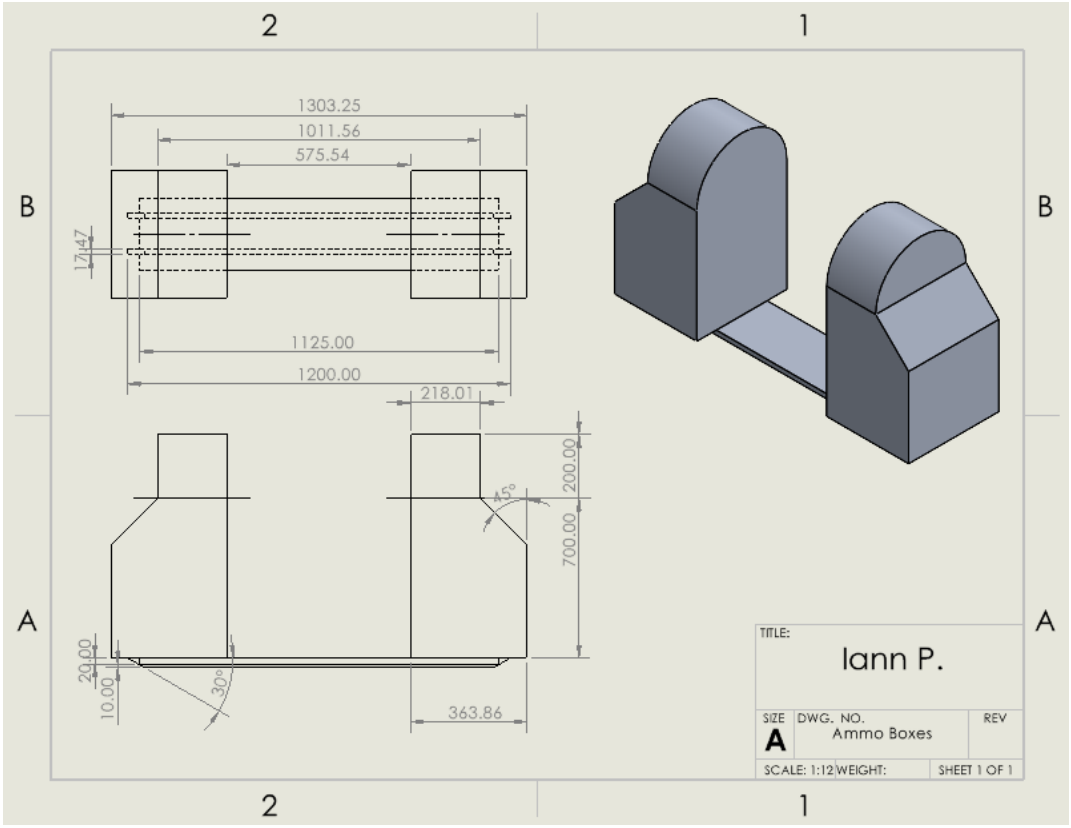

2
1
B
A
112.30
80.00
791.76
942.21
34.13
R380.00
Ø720.00
Ø25.00
195.00
145.00
97.50
R45.89
Ø52.16
ALL FILLETS
R20.00
TITLE:
Iann P.
SIZE
A
DWG. NO.
Gun Wheel
REV
SCALE: 1:16
WEIGHT:
SHEET 1 OF 1


2
1
B
A
Ø370.00
Ø400.00
Ø60.00
Ø60.00
45 deg Camber
250.00
50.00
20.00
Ø360.00
TITLE:
Iann P.
SIZE
A
DWG. NO.
Foot
REV
SCALE: 1:8
WEIGHT:
SHEET 1 OF 1


315.00
10.42
39.47
484.62
R490.00
R500.00
39.57
60.00
10.00
140.00
48.24
6.11
3.81
Ø40.00
Ø120.00
TITLE:
Iann P.
SIZE
A
DWG. NO.
Chair
REV
SCALE: 1:12
WEIGHT:
SHEET 1 OF 1


921.74
TRUE R20.00
200.00
Ø40.00
30.00
Ø660.00
R370.00
250.00
230.00
550.00
R460.00
Ø720.00
80.00
15
550.00
43.87
40.00
80.00
30.00
351.35
920.00
TITLE:
Iann P.
SIZE
A
DWG. NO.
Body Top
REV
SCALE: 1:24
WEIGHT:
SHEET 1 OF 1


2
1
B
A
45°
90°
Ø60.00
R50.00
Ø660.00
Ø740.00
150.00
20.00
13.66
60.00
190.00
200.00
150.00
2137.03
TITLE:
Iann P.
SIZE
A
DWG. NO.
Body (Bottom)
REV
SCALE: 1:32
WEIGHT:
SHEET 1 OF 1


2
1
B
A
1303.25
1011.56
575.54
17.47
1125.00
1200.00
218.01
200.00
45°
700.00
20.00
10.00
30°
363.86
TITLE:
Iann P.
SIZE
A
DWG. NO.
Ammo Boxes
REV
SCALE: 1:12
WEIGHT:
SHEET 1 OF 1

Analysis

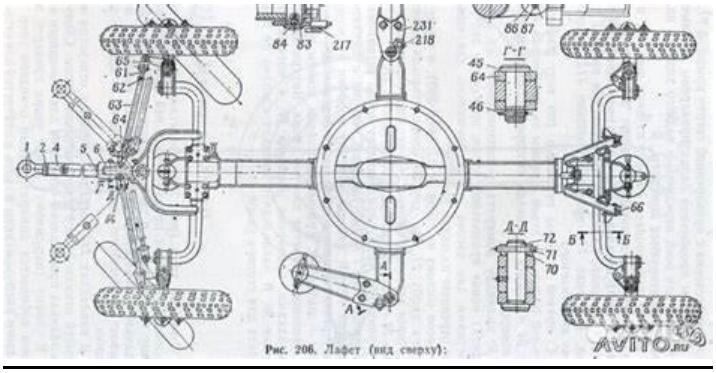
Ultimately what I wanted this project to be able to do is to rotate the AA – gun to look side to side and up and down with the use of the hand cranks on the body (top). I also learned what kind of gear the ZPU – 4 uses to rotate side to side and up and down. It is called the worm and worm gear. I had trouble trying to figure out how to put it into this project, so I just made a gear mate with the Gun Wheel to the hand cranks and similarly to the Body (bottom) to the hand crank.

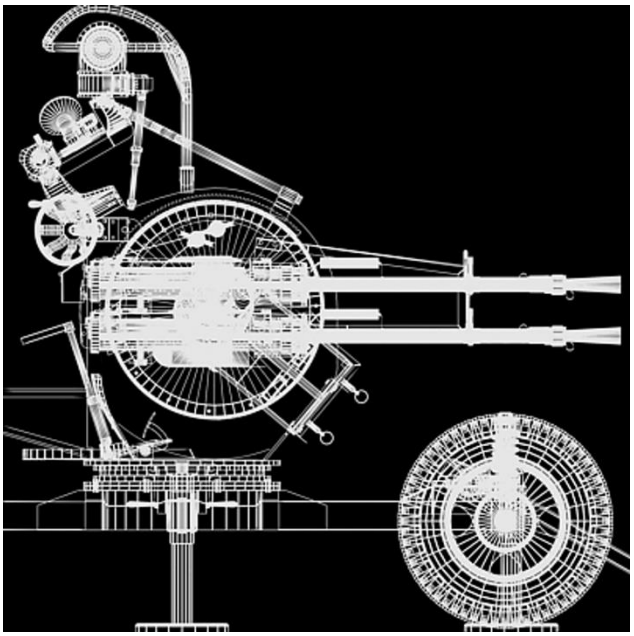
Conclusion:

This was a fun project to take on and a fun class. Although there were many problems with doing some features on the project here, I had the patience to learn and back track to figure out where the root of the problem came from. Something I did learn was how to put planes in the middle of a circular surface, it was with a point and a reference plane on the point and the edge of the circular surface. If I had more practice and knowledge of SolidWorks, maybe there could have been an easier way to model this project, because going off dimensions for a machine gun and making a 3d plane with referenced photos was not easy. The dimensions are most likely not accurate to the real life ZPU-4, but I could say it closely resembles one at most.

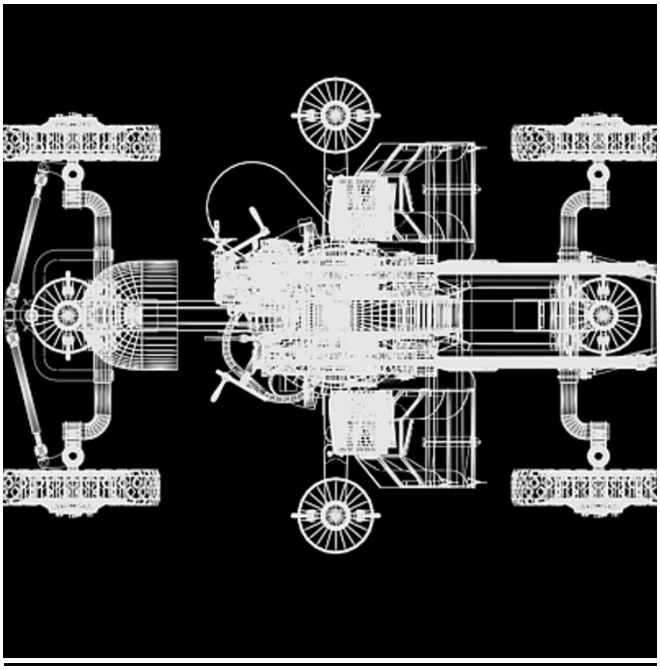
Appendix:

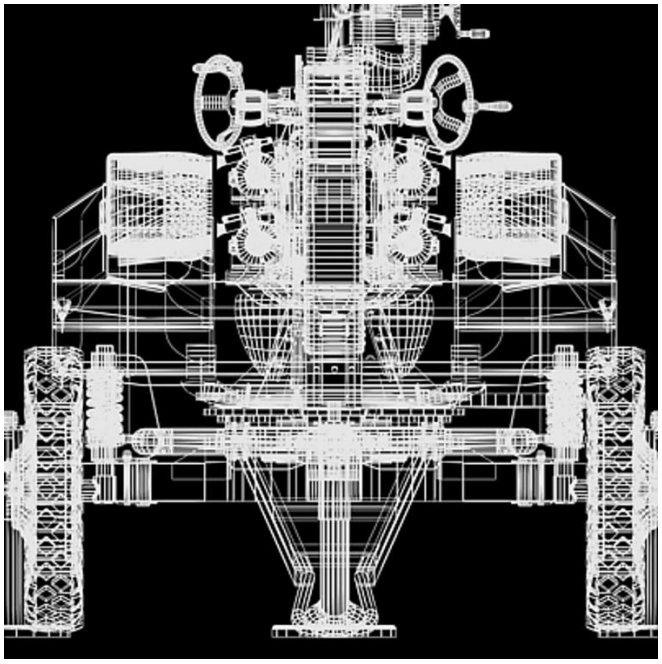
Reference photos I used to create components.













These websites helped me the most:

3d model soviet zpu-4

3D file ZPU-4 🔫 ・Model to download and 3D print・Cults (I did not buy the 3D file, too expensive)

need blueprint - ZPU-4&ZPU-2 (USSR AA-MG)

SolidWorks Commands used sheet

Features Checklist

A minimum of 10 different commands should be used from the following table

Put a checkmark next to the features that you used

Boss/Base
Extrude..
Revolve..
Sweep..
Loft..
Boundary..
Thicken..
Cut
Extrude..
Revolve..
Sweep..
Loft..
Boundary..
Thicken..
With Surface..
Features
Fillet/Round..
Chamfer..
Hole
Simple..
Wizard..
Draft..
Shell..
Rib..
Scale..
Dome..
Freeform..
Deform..
Indent..
Flex..

Pattern/Mirror
Linear Pattern..
Circular Pattern..
Mirror..
Curve
Split Line..
Projected..
Composite..
Curve Through XYZ Points..
Curve Through Reference Points..
Helix/Spiral..
Reference Geometry
Plane..
Live Section Plane..
Axis..
Coordinate System..
Point..
Center of Mass..
Mate Reference..
Grid System..
Tables
Bill of Materials..

Any other features used for your work
Sketch Picture

Estimate number of features used in the project: 15